

HAJI ISMAEL IBRAHIM

Software Engineer • Backend & Distributed Systems

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SUMMARY

Backend and Distributed Systems Engineer with core expertise in Go concurrency, socket programming, and low-latency database architecture. Proven track record shipping production-grade services, including a high-concurrency forum handling 100+ active users and a three-tier search engine using SQLite FTS5, Redis, and LLM fallbacks. Drives engineering standards through Docker workflows, GitHub Flow code reviews, and peer technical mentorship.

TECHNICAL SKILLS

Languages: Go, Python, SQL (PostgreSQL), JavaScript, HTML/CSS

Backend & Frameworks: FastAPI, Flask, Spring Boot, Deno/Node/React, REST APIs

Data & Infrastructure: PostgreSQL, MongoDB, Redis, SQLite (FTS5), Docker/Podman, AWS, Linux

Concepts & Practices: Distributed systems, Concurrency, Socket programming, CI/CD (GitHub Actions), Code review, Git/GitHub Flow, Mentorship

EXPERIENCE

Zone01 Kisumu

Software Engineering Apprentice

Jul 2025 – Present

Kisumu, KE

- Architected a high-concurrency Go forum supporting 100+ concurrent users and a net-cat network utility handling up to 10 simultaneous connections, using socket programming and idiomatic Go concurrency patterns.
- Implemented graph and search algorithms (Edmonds-Karp maximum flow, Backtracking, and Depth-First Search) for optimized routing and problem-solving.
- Built hackathon prototypes: "Dawatrace," a Solidity/Polygon pharma-provenance platform with gasless QR verification; "Agrismart," an offline-capable local-first PWA; "ProperSats," a Bitcoin-powered multi-sig land-listing.
- Led technical Lightning Talks and mentored peers through rigorous code reviews under GitHub Flow, using Docker and SQLite.

Mynt

Project Assistant

Aug 2023 – Sep 2024

Nairobi, KE

- Programmed a secure cryptographic digital-signature utility to automate authentication of internal corporate documentation; led a Figma-based UX redesign of the Mynt mobile application.

Upwork & Fiverr

Freelance Software Engineer

May 2020 – Jul 2023

Remote, KE

- Delivered production-ready REST APIs and full-stack features for a couple of international clients using Python (Flask), Go, and JavaScript (React).
- Integrated third-party APIs, authentication systems, and payment gateways; audited legacy codebases to fix performance bottlenecks and security vulnerabilities.

PROJECTS

Kamugram | Swahili Search Engine & Telegram Bot

github.com/hajisml/kamugram

Technologies: *Python, aiogram, SQLite (FTS5), Redis, Podman Compose, Gemini AI*

- Implemented a three-tier fallback search pipeline that checks local SQLite indexes first, queries live web sources second, and falls back to Gemini LLM for missing definitions.
- Cut query response times by 80% by building an asynchronous Redis caching layer (aiohhttp) on top of 16,000+ indexed entries.
- Configured containerized deployment pipelines using Podman Compose to manage asynchronous task queues and persistent volume state in Linux environments.

VoltPass | Telegram Credential Manager

github.com/hajisml/voltpass

Technologies: *Python, MongoDB, HTTPS, Linux*

- Designed an asynchronous HTTPS webhook architecture to process real-time POST payloads from Telegram API endpoints securely.
- Engineered a MongoDB CRUD engine with strict data isolation, single-click payload copying, and administrative telemetry dashboards.

EDUCATION

KCA University

B.Sc. Software Development | *Second Class Honours (Upper Division)*

Nov 2023

Nairobi, KE

ADDITIONAL

Languages: English & Swahili (native), Intermediate Arabic **Interests:** Workstation tooling & user-space sovereignty, Linux customization, Basketball